INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

TRAILER, MUNITIONS, SMALL (MHU-126 A/M)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a munitions trailer designated as MHU-126 A/M.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1740

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specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solication (see 6.2).

SPECIFICATIONS

FEDERAL

VV-L-800 - Oil, Lubricating, Water Displacement

DEPARTMENT OF DEFENSE

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Combat/tactical Service
MIL-F-3541 - Fittings, Lubrication
MIL-PRF-10924 - Grease, Automotive and Artillery
MIL-V-62038 - Vehicles, Wheeled: Preparation for Shipment and Storage of

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualifications
MIL-STD-410 - Non-Destructive Testing Personnel Qualification and Certification
MIL-STD-831 - Test Reports, Preparation of

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents , drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOCUMENTS

NAVSEA 0900-LP-003-8000 Surface Inspection Acceptance Standards for Metals

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DRAWINGS

Naval Air Systems Command DL 804AS100-5, Data List Trailer, Munitions, Small

(Copies may be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM - E1444 - Particle Examination, Magnetic (DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19248-2959.)

AMERICAN WELDING SOCIETY (AWS)

AWS-D1.1 - Steel Structural Welding Code, Standard for (DoD Adopted)

(Applications for copies should be addressed to the American Welding Society, 550 NW LeJeune Road, Miami, Florida 33126.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Construction. The trailer shall be manufactured in accordance with data list DL 804AS100-5 and all documents listed thereon.

3.1.2 Welding. All welding shall be in accordance with AWS- D1.1.

3.1.2.1 Welders. Welding shall be performed by personnel qualified under MIL-STD-248.

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3.1.2.2 Welding Inspection. During fabrication, all welding shall be inspected in accordance with ASTM-E1444 by personnel qualified in accordance with MIL-STD-410.

3.1.3 Lubrication.

3.1.3.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541.

3.1.3.2 Lubrication. The trailer shall be lubricated with grease in accordance with MIL-PRF-10924 or lubricating oil in accordance with MIL-L-2104 or VV-L-800, as applicable.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with the requirements of this document, the drawings and the contract (see 4.2).

3.2.1 First article sample. A first article sample shall be manufactured by the same procedures and processes, and at the same location, as proposed for the balance of the contract production.

3.3 Performance requirements and product characteristics.

3.3.1 Proof load. The trailer, when supported by its running gear, shall withstand a proof load of 15,000 pounds when tested in accordance with 4.5.5.1.

3.3.2 Brake system.

3.3.2.1 Parking brakes. The parking brakes shall be capable of holding the trailer, loaded with a 5,000 pound test load, on a 10° (17.6 percent) grade in either direction, front or rear.

3.3.2.2 Service brakes. The service brakes shall be capable of stopping the trailer, loaded with a 5,000 pound test load, within 30 feet when traveling at 20 miles per hour, on level, dry concrete.

3.3.3 Towing force. The towing force required to move the trailer from rest on a smooth, dry, level, paved surface, such as brushed concrete, shall not exceed 50 pounds per ton of maximum gross weight, measured at the drawbar.

3.4 Mobility. The trailer shall transport a variety of munitions from magazine to aircraft over the roads as described in table II. Adapters shall be used to accommodate the different munitions. Road and load test(s) (see 4.5.5 and 4.5.6) shall verify transport capability.

3.5 Surface inspection of metals. Non-destructive inspection shall be made of all metallic parts. This inspection shall verify that the surface conditions are in accordance with Class 1 of NAVSEA 0900-LP-003-8000.

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3.6 Marking. Marking for shipment shall be accordance with MIL-STD-129.

3.7 Workmanship. The trailer, including all parts and accessories, shall be fabricated to ensure appearance, fit and adherence to specified tolerances. The trailer shall meet the performance requirements, (see 3.3 and 3.4).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2),

b. Conformance inspection (see 4.3).

c. Sampling inspection (see 4.4).

4.2 First article inspection. The first article sample (see 3.2.1) shall be subjected to the tests noted in Table I under First Article.

TABLE I. Inspections.

Test	Requirement	Test Paragraph	First Article	Conformance
Examination of Product	3.1.1	4.5.1 4.5.1.1	X	X
Conformance to Drawing	3.1.1	4.5.2	X	X
Parking Brake	3.3.2.1	4.5.3.1	X	1/
Service Brake	3.3.2.2	4.5.3.2	X	1/
Mobility	3.4	4.5.6	X	
Overload	3.3.1	4.5.5	X	1/
Towing Force	3.3.3	4.5.4	X	

1/ One trailer of each 25 trailers produced shall be subjected to tests noted.

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4.3. Conformance inspection. Each trailer shall be subjected to the tests in table I under Conformance.

4.4 Sampling inspection. Trailer components shall be inspected using sampling plans as noted on the detail drawings or if none is specified as directed by the procuring authority. These sampling inspections shall be part of the conformance inspection.

4.4.1 Rejection and retest. When a unit fails to satisfy the tests noted in 4.2 and 4.3, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined and resolved.

4.4.2 Defects in items already accepted. In items already accepted, the contractor shall advise the procuring activity of all defects to be found and methods of correcting them.

4.5 Test methods

4.5.1 Examination of products. Each trailer shall be examined visually to determine compliance with the requirements of this specification for which tests are not specified, including materials, and workmanship. Particular attention shall be given to routing, protection, and functioning of the lighting circuits and brake lines.

4.5.1.1 Functioning. Each trailer shall be examined to determine that wheels are correctly aligned, that all brake systems function including back up, and are correctly adjusted; that the electrical system is in proper working order; that all points requiring lubrication are properly lubricated; and that wheel bearings are properly adjusted and lubricated.

4.5.2 Conformance to drawings. Trailers shall be inspected to verify conformance to the engineering drawings.

4.5.3 Brake system tests.

4.5.3.1 Parking brake tests. The trailer, loaded with a 5,000 pound test load (see 4.5.5.2), shall be placed on a 10° inclined ramp, facing uphill, and with parking brakes applied the trailer shall not roll. The loaded trailer shall then be tested facing downhill.

4.5.3.2 Service brakes. The trailer, loaded with a 5,000 pound test load (see 4.5.5.2), shall be connected to a prime mover. The service brakes of the trailer shall be tested in the following manner.

4.5.3.2.1 Brake stopping capability test. Tow the loaded trailer at a speed of 20 mph on a level, dry concrete surface. Stop by applying brakes of towing vehicle thus actuating the trailer inertia brake. The trailer shall stop within 30 feet or less from the point where the brakes were first applied. The trailer shall exhibit no tendency to jack-knife. At the end of the fifth cycle,

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inspect the service brakes and actuating mechanism. Failure to operate or evidence of excessive heating or distortion shall be cause for rejection.

4.5.3.2.2 Back-Up test. After acceptance under 4.5.3.2.1, the trailer shall be subjected to a back-up force. The service brakes shall momentarily hold but with continued force shall release and allow unrestricted rearward travel.

4.5.4 Towing force test. A scale shall be placed between the prime mover and the towbar of the trailer, loaded with a 5,000 pound test load (see 4.5.5.2). The prime mover shall apply a gradual towing force through the scale to the loaded trailer, with its brakes released and with the trailer on level concrete. The trailer shall roll before the scale registers 175 pounds.

4.5.5 Load tests.

4.5.5.1 Proof load test. The proof load testing shall be accomplished by placing a 15,000 pound test load across two steel bars which have been placed across the trailer rails at hole locations 7 and 46 as shown on the detail drawings to determine conformance to 3.3.1.

4.5.5.2 5,000 pound test load. The 5,000 pound test load shall consist of bulk material such as steel or concrete which, when placed across two steel bars located across hole locations 7 and 46 (as shown on the detail drawings) shall not restrict the trailer cramping angle or drawbar movement.

4.5.6 Mobility tests. During the following tests, the trailer shall be closely observed for its ease of handling. Any tendency to yaw, sway, dog walk or jack-knife shall be cause for rejection.

4.5.6.1 Road tests. The trailer loaded with a 5000 pound test load shall be subjected to the following table II road tests:

TABLE II. Road tests.

Surface	Speed (mph)	Distance (Miles)
Level paved highway	15-20	100
Graded gravel road	5-10	50
On a level course with one inch chains installed on all tires	5-10	5

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4.5.6.2 Maneuverability. Upon completion of road testing (see 4.5.6.1), the loaded trailer shall be towed over level paved highway at speeds of 8 mph through 50 right circle turns and 50 left circle turns both at maximum cramping angle. It shall then perform 25 sudden stops at speeds of 20 mph. At the conclusion of these tests and those described in 4.5.6.1, the vehicle shall be closely examined. Any evidence of excessive wear or damage shall be cause for rejection.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and marking shall be in accordance with the requirements of MIL-V-62038.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The MHU-126 A/M Small Munitions Trailer is a shorebased version of the MHU-191/M Weapons Skid and has been designed for use with a variety of adapters currently used with MHU-191/M to transport munitions from magazine to aircraft.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specification issue of individual documents referenced (see 2.2.1).

c. Whether first article inspection is required. (See 3.2)

d. Packaging requirements (see 5.1)

6.3 First article. When first article inspection is required, the item to be tested should be a first article sample. The first article should consist of one unit. The contracting officer should

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include specific instructions in the acquisition documents regarding arrangements for examinations, tests and approval of the test results. Further information about first article requirements can be obtained from Commander, Naval Air Warfare Center Aircraft Division, Code 4826, MS562-3, Highway 547, Lakehurst, NJ 08733-5130.

6.4 Subject term (key word) listing.

Ammunition
Armament
Bullets
Ordnance
Vehicle

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy - AS
(Project 1740-0180)

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